Feature selection and data leakage prevention

"Feature selection and data leakage prevention in clinical or public health prediction models using survey data: multicollinearity (VIF), statistical filtering and model-based importance. Return a matrix with: authors, year, journal, DOI, data type, selection methods used, and effect on model performance. Limit to 2015-2026." *Generated by LeapSpace*

Feature selection and leakage control in clinical/public health prediction models

The clearest pattern across the retrieved studies is that **survey-based clinical and public health models usually combine statistical filtering with redundancy control and, increasingly, model-based selection**, while **data leakage prevention is handled by doing all filtering inside the training data or by explicitly removing post-outcome variables** [1] [2] [3] [4] [5] [6].

Authors	Year	Journal	DOI	Data type	Selection methods used	Effect on model performance
Not provided in retrieved evidence; study on community health survey / credit card use data	2023	Frontiers in Public Health	10.3389/fpubh.2023.1257861	Public health survey data	VIF screening at a relaxed threshold of 30; ordinary least squares filtering with $p > 0.05$ removed; features then used in ML models [7]	Reduced multicollinearity without removing key factors; the selected feature sets were used to build obesity models, but the excerpt does not report a single global performance delta versus an unfiltered baseline [7].
Not provided in retrieved evidence; health survey-based chronic child malnutrition model (Peru ENDES)	n.d.	Uploaded document	Not stated in retrieved evidence	Large-scale survey microdata	Four-stage selection: remove high-missingness variables; drop collinear variables using Pearson/Spearman correlation and VIF > 5; retain statistically significant variables via chi-square/Kruskal-Wallis; then use gradient-boosted tree importance to choose final predictors [8]	The feature pipeline reduced 189 raw columns to 45 predictors; the final LightGBM model achieved AUC-ROC 0.8297 and retained the same performance after retraining on the 45-variable set [8] [9].
Not provided in retrieved evidence; same Peru ENDES model	n.d.	Uploaded document	Not stated in retrieved evidence	Survey microdata	Leakage prevention: excluded child height, raw weight, and any post-measurement treatment/diagnostic variables; all preprocessing and selection were done before modeling [6]	This was a methodological safeguard rather than a performance booster; the study reports that excluding leakage-prone variables preserved validity and avoided spuriously inflated performance [6].

Not provided in retrieved evidence; same Peru ENDES model	n.d.	Uploaded document	Not stated in retrieved evidence	Survey microdata	Model-based importance with gradient boosting; SHAP used after model fitting for interpretability; no separate leakage-prone features were kept in the final pipeline [10] [11] [12]	LightGBM was selected as best among six algorithms, with AUC-ROC 0.8297 and F1 49.3%; re-training on the 45 final variables did not change performance, suggesting the selection step preserved predictive signal while shrinking dimensionality [9].
Not provided in retrieved evidence; machine learning on health survey data	2025	Alexandria Engineering Journal	10.1016/j.aej.2025.11.050	Health survey data	RFECV applied only within training folds; LightGBM used as the base estimator; wrapper selection based on model importance rather than univariate filtering [2]	The workflow identified 20 features as the best balance between complexity and generalization; the excerpt frames this as improving the model's robustness and avoiding overfitting, but does not provide a numeric performance gain in the snippet [2].
Not provided in retrieved evidence; mammographic + clinical data	2026	Frontiers in Oncology	10.3389/fonc.2026.1762400	Clinical and imaging data	Univariate statistical filtering as primary method; LASSO and RFE used in sensitivity analyses with largely overlapping subsets [3]	The study chose univariate filtering for interpretability and to reduce overfitting risk in a moderate-size dataset; the alternative methods produced similar feature sets, implying selection robustness rather than a major performance divergence [3].
Not provided in retrieved evidence; child health survey / stunting prediction	2024	2024 9th International Conference on Informatics and Computing, ICIC 2024	10.1109/ICIC64337.2024.10956483	Public health survey data	Filter methods: mutual information, Pearson correlation coefficient, and information gain; 37 of 84 variables retained [4]	Feature selection produced small performance changes: RF improved from 0.84600 to 0.85701 accuracy and GB from 0.86330 to 0.86124; the main gain was dimensionality reduction and efficiency, not a large accuracy jump [4].
Not provided in retrieved evidence; elderly T2DM readmission model	2025	Frontiers in Endocrinology	10.3389/fendo.2025.1534516	Clinical data	Relevance-based feature selection; medication variables aggregated to broader categories; VIF checked and all selected variables had VIF < 5 [13]	The excerpt reports improved accuracy and runtime after removing irrelevant features, but does not give the exact delta in the snippet; the main effect was a more robust and generalizable model [13].
Not provided in retrieved evidence; postoperative liver cancer resection model	2026	BMC Surgery	10.1186/s12893-026-03746-x	Clinical cohort data	"Triangulated" selection: LASSO + XGBoost RFE + Boruta; strict intersection of the three; all steps done in training only to prevent leakage [1]	The combined approach was explicitly intended to reduce multicollinearity, avoid single-method bias, and prevent leakage; the snippet does not report a numeric gain, but states the final model was evaluated on a held-out test cohort [1].

Bottom line

Across these studies, **VIF-based screening is mainly used to remove redundant predictors, univariate/statistical filters are used to keep interpretable signals, and model-based importance methods (RFECV, gradient boosting, Boruta, SHAP-guided ranking) are used when the goal is to preserve nonlinear interactions** [1] [2] [4] [7] [8] [12]. The strongest leakage-prevention practice is **restricting feature selection to the training split and excluding any variables measured after the outcome or that directly encode it** [1] [6]. Performance effects are usually **modest but positive or neutral**: in the Peru survey model, the final 45-feature set preserved AUC-ROC at 0.8297, while in the stunting study feature selection slightly improved RF and left GB essentially unchanged [4] [9].

Confidence level: **Medium** — the evidence is strong for the methods and leakage controls, but several retrieved excerpts do not provide author names or full quantitative performance deltas, so the matrix is complete only to the extent supported by the snippets [1] [2] [3] [4] [6] [7] [8] [9] [13].

References

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